

Smart Medication Management System

Overview

Smart Medication Management System is a healthcare-focused web application that helps users manage their medications, schedules, and medication adherence. The system enables users to create medication reminders, track medication intake, and monitor adherence through an analytics dashboard.

Problem Statement

Many individuals, especially senior citizens and busy professionals, forget to take medications on time. Missing medications can lead to health complications and reduced treatment effectiveness. Existing solutions often lack simplicity, tracking capabilities, or personalized reminders.

Solution

The Smart Medication Management System provides:

- User Authentication
- Medication Management
- Schedule Management
- Reminder Tracking
- Medication Adherence Monitoring
- Analytics Dashboard

Users can register, add medications, create schedules, receive reminders, and mark medications as taken.

Features

Authentication

- User Registration
- User Login
- Session Management

Medication Management

- Add Medication
- View Medication
- Edit Medication
- Delete Medication

Schedule Management

- Create Reminder Schedule
- View Schedule
- Edit Schedule
- Delete Schedule

Medication Tracking

- Mark Medication as Taken
- Daily Medication Logging

Analytics Dashboard

- Total Medications
 - Total Schedules
 - Medications Taken Today
 - Adherence Percentage
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Technology Stack

Frontend

- Streamlit

Backend

- FastAPI

Database

- SQLite

ORM

- SQLAlchemy

Background Processing

- APScheduler
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Database Design

Users

- id

- name
- email
- password

Medications

- id
- user_id
- name
- dosage

Schedules

- id
- medication_id
- reminder_time

Medication Logs

- id
 - schedule_id
 - date
 - taken
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Project Workflow

1. User Registers/Login
 2. User Adds Medication
 3. User Creates Schedule
 4. Reminder Appears on Dashboard
 5. User Marks Medication as Taken
 6. System Stores Medication Log
 7. Analytics Dashboard Updates
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Future Enhancements

- Google Calendar Integration
 - Email Notifications
 - SMS Notifications
 - WhatsApp Reminders
 - JWT Authentication
 - PostgreSQL Migration
 - Cloud Deployment
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Learning Outcomes

- REST API Development
- FastAPI Backend Development
- SQLAlchemy ORM
- SQLite Database Management
- Frontend Development with Streamlit
- CRUD Operations
- Authentication Systems
- Healthcare Workflow Design